

GPU Application

CUDA Application

NetGPU Frontend

Parse PTX

→ CTX

→ LLVM / MLIR

Portable GPU IR

CTX IR (Concordi Thread eXecution)

Portable GPU IR with Metadata for State Management

JIT Compilation

CTX → PTX

CTX → ROCm

CTX → SPIR-V

CTX → TOSA

Runtime System

Abstraction Layer: Thread Mgmt

Memory Ops

Synchronization

SIMT↔MIMD

Hardware Targets

NVIDIA GPUs

H100 / A100 (SIMT)

AMD GPUs

MI250 / RX

Intel GPUs

Arc / Xe

Future Targets...